

Information
Theory and
Recursive
Meta-
Systemic
Data
Specification

Based on 49 sources

Term/Concept	Definition/Description	Mathematical Formulation	Source/Origin
Entropy	A measure of uncertainty or information content, representing the average amount of information produced by a source or the "missingness" in a signal.	$H = -\sum p_i \log p_i$	Claude Shannon
Channel Capacity	The maximum rate at which information can be transmitted reliably over a channel or learned from evidence.	$C = \max_{p(x)} I(X; Y)$	Claude Shannon
Equivocation / Cohomological Gaps	The measure of information loss or missingness; remapped as structural voids/defects where local data cannot be patched into a global conclusion.	$H(X Y)$ or $H^k = \ker(\partial)/\text{im}(\partial)$	Claude Shannon / HALIRA Framework
Torsion	An algebraic or geometric invariant representing the resistance of a local system to integration; physicalized as cognitive friction from paradoxes	$\tau \in H^n(X; G)$ or $G = \text{spike}(\nabla^6 \Phi)$	HALIRA / Higher Topos Theory